

VISVESVARAYA TECHNOLOGICAL UNIVERSITY
“JNANA SANGAMA”, BELAGAVI - 590 018



A PROJECT PHASE - I REPORT

on

“Monitoring System for Street Light Fault Detection
And Location Tracking”

Submitted by

Brinda R Shetty

4SF22IS022

Prathik Poojary

4SF22IS074

Shivani K

4SF22IS096

Vaishnav L

4SF22IS122

In partial fulfillment of the requirements for the VI semester

of

BACHELOR OF ENGINEERING

in

INFORMATION SCIENCE & ENGINEERING

under the guidance of

Mrs. Prajna U R

Assistant Professor

Department of ISE

at



SAHYADRI

College of Engineering & Management

An Autonomous Institution

Adyar, Mangaluru - 575 007

AY: 2024 - 25

SAHYADRI
College of Engineering & Management
An Autonomous Institution
Adyar, Mangaluru - 575 007

Department of Information Science & Engineering



CERTIFICATE

This is to certify that the Project Work Phase I has been carried out by **Brinda R Shetty (4SF22IS022)**, **Prathik Poojary (4SF22IS074)**, **Shivani K (4SF22IS096)** and **Vaishnav L (4SF22IS122)**, the bonafide students of Sahyadri College of Engineering & Management, in partial fulfillment of the requirements for the VI semester of **Bachelor of Engineering in Information Science & Engineering** of Visvesvaraya Technological University, Belagavi, during the Academic Year 2024-25. It is certified that all corrections/suggestions indicated for Continuous Internal Assessment have been incorporated in the report deposited in the departmental library of Information Science & Engineering. The Project Work Phase I report has been approved as it satisfies the academic requirements in respect of project work prescribed for the said degree.

Project Guide
Mrs. Prajna U R
Assistant Professor
Dept. of ISE, SCEM

Project Coordinator
Mr. Vasudeva Rao P V
Assistant Professor
Dept. of ISE, SCEM

HOD
Dr. Rithesh Pakkala P.
Associate Professor & Head
Dept. of ISE & CSE(DS), SCEM

SAHYADRI
College of Engineering & Management
An Autonomous Institution
Adyar, Mangaluru - 575 007

Department of Information Science & Engineering



DECLARATION

We hereby declare that the entire work embodied in this Project Phase - I Report titled **“Monitoring System for Street Light Fault Detection And Location Tracking”** has been carried out by us at Sahyadri College of Engineering and Management, Mangaluru, under the supervision of **Mrs.Prajna U R** in partial fulfillment of the requirements for the VI semester of **Bachelor of Engineering in Information Science & Engineering**. This report has not been submitted to this or any other university for the award of any other degree.

Brinda R Shetty (4SF22IS022)

Prathik Poojary (4SF22IS074)

Shivani K (4SF22IS096)

Vaishnav L (4SF22IS122)

Dept. of ISE, SCEM, Mangaluru

Abstract

This project presents the design and implementation of an Smart Streetlight Monitoring System that utilizes modern embedded technologies to optimize energy consumption and streamline urban lighting management. The system integrates microcontrollers to process inputs from various sensors that detect ambient light and motion, enabling automatic control of streetlights based on environmental conditions. A GSM module facilitates real-time communication and remote monitoring by transmitting data to a centralized web application, which provides an intuitive interface for users to oversee and manage streetlight performance. Through careful programming and integration, the system enhances energy efficiency, reduces manual intervention, and supports scalable deployment, contributing to the development of sustainable smart cities.

Acknowledgement

It is with great satisfaction and euphoria that we are submitting the Project Phase - I Report on “**Monitoring System for Street Light Fault Detection And Location Tracking**”. We have completed it as a part of the curriculum of Visvesvaraya Technological University, Belagavi in partial fulfillment of the requirements for the VI semester of Bachelor of Engineering in Information Science & Engineering.

We are profoundly indebted to our guide, **Mrs.Prajna U R**, Assistant Professor, Department of Information Science & Engineering for innumerable acts of timely advice and encouragement.

We also thank **Mr. Vasudeva Rao P V**, Project Coordinator & Assistant Professor, Department of Information Science & Engineering for the constant encouragement and support extended throughout.

We express our sincere gratitude to **Dr. Rithesh Pakkala P.**, Assoc. Professor & Head, Department of ISE & CSE(DS) for his invaluable support and guidance.

We sincerely thank **Dr. Shamantha Rai B**, Dean Academics, and **Dr. Sudheer Shetty**, Vice Principal, Sahyadri College of Engineering & Management, for their constant encouragement during the project work.

We sincerely thank **Dr. S S Injagneri**, Principal, Sahyadri College of Engineering & Management, who has always been a great source of inspiration.

Finally, yet importantly, we express our heartfelt thanks to our family & friends for their wishes and encouragement throughout the work.

Brinda R Shetty (4SF22IS022)

Prathik Poojary (4SF22IS074)

Shivani K (4SF22IS096)

Vaishnav L (4SF22IS122)

Table of Contents

Abstract	i
Acknowledgement	ii
Table of Contents	iii
List of Figures	iv
List of Tables	v
1 Introduction	1
2 Literature Survey	5
3 Problem Statement	21
4 Requirements Specification	22
5 Proposed Methodology	24
5.1 Overview	24
5.2 System Architecture	24
5.3 Step-by-Step Methodology	24
5.4 Advantages of the Proposed Methodology	26
6 Expected Outcomes	27
7 Project Plan	29
8 Conclusion	31
References	33

List of Figures

5.1	System architecture of the smart street lighting system	25
7.1	Gantt Chart	30

List of Tables

4.1	Software Requirements	22
4.2	Hardware Requirements	23

Chapter 1

Introduction

In modern-day urban space, the quality of public infrastructure plays a fundamental role in the daily lives of users. Of the myriad contributors to urban infrastructure, street lighting is very possibly the most important. Accessible street lighting promotes safe movement when it is dark, reduces the potential of road accidents, and provides a feeling of comfort and safety to pedestrians and drivers alike. Adequate street lighting can also help to beautify urban spaces, and promote socializing and paramedic activity after dark. Even with the fundamental importance to the use and enjoyment of urban spaces, streetlight systems are largely unnoticed, poorly managed, and rarely maintained, often relying on automatic decision-making systems that rely on outdated and inefficient manual analysis and processes. "Streetlight management" involves slow, inefficient manual processes that require resources and always have the potential for human error.

Traditionally, municipal authorities deal with streetlight failures in one of two ways, either through a routine manual inspection of streetlights or by relying on the general public to report failures. This reactive process has many downsides. A streetlight can be out for days or weeks without notice, especially in less dense or lower income neighborhoods which have limited reporting of faults. During this time, residents in these neighborhoods face even higher risks of accidents, theft, or assault in the absence of lighting. These failures not only compromise safety but also deteriorate trust in local government. As well, without automated tracking, municipal staff have trouble coordinating their efforts, responding in a timely manner, as well as incurring unnecessary expenses for maintenance that could be avoided had the field staff been deployed properly.

In light of the challenges faced by public service delivery, the development of smart city initiatives allows for the opportunity to use technology to develop solutions. A smart

city uses digital technologies, data and analytics, and automation, which can improve efficiency, reliability, and sustainability in urban systems. Public lighting systems are integral to urban public service delivery, and smart streetlighting systems can transform the way in which urban public services manage public lighting. Smart streetlighting systems take advantage of microcontrollers, sensors, communication modules, and a centralized data platform through which organizations can gain visibility over the health of streetlights, increase visibility over their performance, proactively detect faults, and reduce downtime to operationalize predictive maintenance, without the need for regular inspections of the streetlights.

This project presents a Smart Streetlight Failure Detection and Monitoring System, an innovative approach to modernizing streetlight failure monitoring and reporting, and associated processes. The design of the proposed system integrates a distributed architecture; each streetlight contains a small microcontroller unit. The microcontroller is connected to various sensors, which monitor distinct parameters including the current flow, voltage levels, and lamp state. Upon identifying a mismatch (for example, a lamp that does not turn on, low current flow, or an interrupted power supply), the system will automatically classify this as a fault. The identified fault is then communicated to a host server via a GSM (Global System for Mobile Communication) module, which enables real-time fault reporting without human interaction.

The centralized web-based platform of the system is the core component—it's the main interface for both administrators and technicians in the field. Administrators can maintain an overview of every monitored streetlight in the system on one dashboard. Faults are reported as a fault with all the details surrounding the fault (e.g., where it was found; when it happened; what type of fault). This allows administrators to prioritize their workload, pass the repairs onto maintenance teams, and keep track of whether and when a fault is repaired. Technicians are able to log into the system to see the repair jobs assigned to them, to update the system to show the status of repairs being completed, and have access to historical fault data to aid them in their decision making.

One unique aspect of the system is that it not only tracks faults in real time, but it also keeps a history of every fault with metadata such as the date, time and duration of the fault and the fault type. Over time this data can be used to perform broad fault/disruption analysis and glean statistics on fault behavior or patterns. For example, if a street or transformer zone is frequently showing faults and therefore disruptions,

the authorities can investigate how to fix the issues underlying that area. This type of analysis could also be beneficial when looking at predictive maintenance algorithms to improve the operational reliability of the lighting network for future failures.

Moreover, the system's GIS-based mapping capabilities enable visual tracking of faults across the city. Faults can be displayed on a digital map using colored markers, allowing maintenance teams to plan the most efficient repair routes and allocate resources strategically. This is particularly beneficial in large cities where logistical planning is crucial to minimizing fuel usage and reducing repair times. The integration of mobile applications for technicians can further streamline the process, allowing real-time updates from the field and enhancing communication between departments.

In addition to technical benefits, the use of control systems creates significant economic and environmental benefits. Automated monitoring and control systems greatly reduce manual/regular inspection needs thereby minimizing labor costs and increasing availability of staff for other important activities. Energy waste is also reduced because these systems can help avoid longer outages and optimize repair times. Their evidence assists staff in the planning of repair programs, which may help reduce energy waste with respect to electricity management. These improvements contribute to reducing the carbon footprint of municipal activities, and promote desired environmental sustainability.

The system is also scalable and can be adapted. The implementation can occur in stages, with a pilot installation starting with high priority or high traffic areas of the city, and expanding to city-wide coverage. Other smart city system integration is possible, such as with traffic sensors, emergency services, or pollution monitoring devices. Furthermore, upgrades in the future could incorporate solar smart lights, lights that can adjust brightness dynamically based on ambient light or pedestrian traffic, or machine learning systems that predict faults. The system could also allow for administration through voice interfaces or reference smart assistants for alerts and status updates with voice.

The system can positively impact society by supporting safer communities and increasing good citizen experiences. Citizens will benefit from repair times that will decline, streets that are designed for security lighting, and visibility into what public service departments accomplish and how. The proposed system reinforces a culture of accountability within municipal departments, as there is a record of every fault and every repair action taken. The smart streetlight system allows for evidence-based governance and for municipalities to make better decisions on policy issues and budgets.

In conclusion, the Smart Streetlight Fault Detection and Monitoring System represents a complete and consolidated answer to the persistent challenges of public lighting maintenance. It intelligently integrates a combination of hardware, communication technology, and data analytics to deliver automated and real-time fault detection and improve efficiency. By addressing the serious contexts of safety, maintenance, and energy use, this project contributes to the broad objectives for sustainable and smart city development. This systems implementation represents a further foothold towards modernizing municipal services and illustrates the way in which emergent technologies can be employed to improve urban living and develop connected and resilient futures.

Chapter 2

Literature Survey

Research in smart street lighting systems has progressed rapidly with the growing need for energy efficiency, automation, and public safety. Various approaches have been explored to improve streetlight automation, fault detection, and adaptability using emerging technologies like IoT, machine learning, and renewable energy.

Quattrocchi et al. [1] developed a very affordable structural health monitoring solution and sensor array for street lighting poles leveraging Raspberry Pi-based edge computing and use sensor modules DHT22, GY-302, and MPU-6050 to capture temperature, humidity, light, acceleration, and tilt data. They used a distributed Edge/Cloud architecture to deploy their monitoring solution and, to store the data, the software used InfluxDB; to visualize, the software used Grafana, which allowed real-time monitoring and alerting. Infrastructure for street lighting is imperative for urban and suburban areas, as it exposes citizens to certain levels of safety during periods of darkness and allows for the eventual engagement of activities later than dusk. Street lighting poles themselves have various structural risks, namely corrosion, fatigue, and material and structural defects which may appear suddenly, thereby exposing people and property to potential danger. Current inspections are primarily based on a visual assessment of the pole or non-destructive test methods. While a visual inspection is cheaper than other testing methods, it usually requires an inspector to view and report on many lighting poles in an urban centre; due to the curses of time and cost to complete a full assessment of lighting poles, structural ignorance tends to disproportionately underestimate structural degradation on poles in either a peripheral area of a city or less frequented areas of a city.

Quattrocchi et al. investigated a low-cost IoT-based Structural Health Monitoring (SHM) system for use in providing continuous real-time monitoring of street lighting

poles to help improve maintenance efficiency and enhanced safety. Their system combined a Raspberry Pi 3 B+, and sensors for measuring ambient temperature, humidity, light intensity, pole acceleration, and pole tilt. The system provides information about the mechanical stressors on the pole and the surrounding environmental conditions, contributing to structural deterioration of the pole or imminent failure. An important innovation of their work is their use of affordable and available sensors (e.g., DHT22, GY-302, MPU-6050) and associated open-source software tools (e.g., Raspbian OS, InfluxDB, Grafana) to create a modular and scalable system that can be built into existing urban infrastructures. The device is encased in a weatherproof enclosure, powered via the electrical grid, and transmits data over 4G cellular networks to a cloud/edge computing platform. The platform design allows for local edge processing of data and remote management through desktop or mobile clients (and also customizable alert notifications using Telegram as an example).

The authors describe a full experimental characterization of sensor performance, examining factors controlling each sensor's measurements (temperature, humidity, light intensity, vibration, tilt) and evaluating their accuracy and reliability of measurement. They articulate some of the limitations of the sensors related to calibration, measurement uncertainties, and sensor errors from time to time, emphasizing caution in conducting metrological assessments and metrological evaluations when using low-cost devices. With all the forementioned challenges considered, the system provides enough accurate measurements and timely responses for it to be used in records and monitoring.

Shagun Malhotra et al. [2] developed a street lighting system that enhanced its features using a GSM module to send light status to the main controller. This caused some improvements in the energy efficiency and monitoring for maintenance. However, when problems arose with the GSM communication, it would have eliminated issues if they had used a better and more reliable ESP8266 Wi-Fi module. Shagun Malhotra et al. proposed a GSM-based smart street lighting system, which aimed for an improvement in operational efficiency as well as the ability to detect faults remotely. They designed their street lights to include a GSM module, allowing the light to send operational data back to a central control unit, which would give a timely warning if a light unit was malfunctioning, allowing for swift contact with any site repair team.

A key feature of this system is that it is simple and, therefore, low power. The GSM module was used to send status information using short message services (SMS), meaning it was usable in a region that has no sophisticated internet infrastructure. The SMS

system was especially important in rural or semi-urban areas, where van inspection is required to find faults (Malhotra et al. 2021). The system did have limitations because of the immaturity of GSM technology as messaging delivery failures and deliberate network delays during peak traffic already had potential compromises to the reliability of the system. The authors recognised that GSM was insufficient and proposed the ESP8266 Wi-Fi module, which is a finite, better alternative in stable Wi-Fi conditions. The ESP8266 module may be better in terms of data processing, real-time two-way communication, and cloud capability, which would greatly enhance scalability and performance.

The research firmly established the importance of integrating wireless technologies when developing smart infrastructures. Although the system sought to identify solutions to communication problems the system opened up the opportunity to allow power-reduced techniques through behaviour based usage and scheduling.

Overall, this paper made a contribution to the field by demonstrating that existing mobile technologies could be utilised for smart city applications. The authors successfully demonstrated an automated fault reporting system that could reduce labour time and may signal the start of a more constructed IoT pathway.

K. Y. Rajput et al. [3] developed a street lamp monitoring system that used GSM for remote collection of information from street lights, which allowed 24-7 monitoring and faults detection. The system would be effective in concept but did require some human involvement to maintain this system, thus incurred a high operating costs. Developed by K. Y. Rajput et al., a GSM-based street lamp monitoring and control system was designed to support real-time detection and troubleshooting of faults in street light infrastructure. Their method relied on outfitting each streetlight with GSM modules, which would send operational data (on/off state, current consumption, etc.) to a monitoring unit. This allowed for continuous monitoring, without the burden of physical inspections, which entail much higher labor costs, and will often result in delays before identifying luminaire malfunctions.

The street lamp monitoring system utilized microcontrollers to analyze sensor data and recognize faults including burnt out lamps and broken circuits. After faults were identified, an SMS message was sent to the appropriate maintenance authority allowing for timely repairs to be made and creating legitimate reliability in public lighting systems.

In conclusion, while the model effectively reduces as much downtime of individual luminaires as possible, while improving delivery of service, there are considerable limitations. Continuous dependence of GSM as the communication method, and a lack of

automation for repairs and decision-making means human involvement will always be required. In some areas, and especially in denser urban areas or remote locations, GSM communication may not be entirely dependable or without potential reporting lapses. Notwithstanding these factors, the solution was feasible. The overall project was inexpensive, easily implemented, and provided a first version of remote monitoring in smart streetlighting. The research showed it is valuable to use existing teleph-communications infrastructure to upgrade public service.

The paper represented an important piece of the smart lighting system, and shows how traditional components, like GSM, could be used effectively to monitor and manage faults in large-scale infrastructure.

Chetna Badgaiyan et al. [4] developed a smart lighting system based on ZigBee and PIR, with ZigBee providing the wireless communication between nodes, and PIR sensors controlling operational lights on motion. The smart lighting offered better efficiency yet added complexity to the system which deterred scaling. Chetna Badgaiyan and colleagues created a smart street lighting system with ZigBee wireless communications and Passive Infrared (PIR) motion detectors. The purpose of the smart lighting system was to use energy more effectively by switching streetlights off or on in relation to the real-time human and vehicle occupancy patterns. This framework enabled them to use PIR to detect motion and, when motion was detected, it used ZigBee to communicate between the streetlight nodes to switch the lights on or off in such a way as to meet occupancy patterns.

The unique involvement of this design is its use of distributed control and localized decisions. Rather than relying on a centralized controller, each lamp was semi-autonomous. And the neighboring lamps were able to communicate with each other using ZigBee mesh networking. The distributed nature of control makes the system scalable and fault tolerant. The failure of one light doesn't bring down the entire network.

Energy savings were notable in areas with little or on-off occupancy. The brought savings also extended the lifetime of lighting units as the time they were unnecessarily on was reduced. The wireless nature of ZigBee also reduced the extent of cabling needed in low-density settings, reducing installation and maintenance costs. Nonetheless, the authors identified limitations specific to urban environments. Dense urban infrastructure, where overhead structures and other elements interfered with transmitting ZigBee signals resulted in missed detections or complete failures in communication. As well, the process of scaling up the system in cities with high electromagnetic noise was not easy.

While their study was not without limitations, it provided a case for distributed, sensor-driven lighting systems. Its demonstrated motion-detection coupled with light-weight communication methods like ZigBee produced meaningful efficiencies while allowing public safety.

B. K. Subramanyam et al. [5] devised a solar-powered lighting system replacing incandescent and fluorescent bulbs with LED lights and sensor modules. Although it aligned with sustainability imperatives as an energy-efficient solution, the authors recognized the associated costs and recommended using LDRs (light dependent resistors) to provide efficient, cost-effective control for varying intensities dependent on local weather conditions shifting as identified by the initial authors research study on using solar energy. B. K. Subramanyam and colleagues proposed a solar-powered streetlight solution based on Light Emitting Diode technology with sensor-based automation. The idea was to eliminate the use of incandescent and fluorescent lights (which are wasteful in terms of energy consumption) in favour of LED lights controlled by light-dependent resistors (LDRs), which would modulate brightness based ambient light levels.

The best aspect of this system (the proposal) was its sustainability aspects. The solar panels charged the batteries during the day and held energy for the LED lights at night. The LDRs could further enhance energy use by dimming the light in the early dawn and dusk hours of the day when the lights do not have to be at full brightness. The system is capable of reducing carbon emissions as well as reducing reliance on grid power, which has implications for remote locations as well as places with no electrical grid infrastructure.

The whole system could be constructed in modules and installed without human involvement. However, the capital outlay for solar panels and battery storage systems proved prohibitive to fund for a municipality with budgetary restraints, especially when deploying the system on a city-wide scale. One suggestion offered by the authors was to utilize less expensive LDR sensors as a viable statistical method for lighting regulation without using costly A.I. or cloud management software. The study concentrated on energy efficiency, although it touched upon future scalability relative to smart city requirements. The system could both include a wireless communication module or internet-of-things (IOT) ability for remote fault detection, as well as remote monitoring capabilities.

This study examined the potential of renewable energy in public infrastructure and illustrated an excellent case for combining sustainability and automation when it came to urban lighting systems.

Smita A. Bhosale [6] designed an energy efficient system for modified increasing or decreasing light intensity dependent upon vehicle movement also incorporating atmospheric changes based on communication to the system via ZigBee. The system was responsive to vehicle motion and environmental variables ensuring proper use of lighting systems with energy savings to the local governing authority. During the field operational testing, given the density of the urban zone ZigBee communication was challenged by interference from the surrounding population. Smita A. Bhosale presented an efficient street lighting system that was activated by vehicle/people movement and reactions to different conditions using ultrasonic motion sensors, microcontrollers, and ZigBee wireless communication with two objectives, to reduce wasted energy in systems for public lighting, where lights were only turned on when they were realistically needed - depending on the natural changes in the world outside and the instantaneous detection of movement.

The system architecture was sensor driven in that it used passive infrared (PIR) sensors to detect some type of vehicular or human movement, while several environmental parameter sensors (temperature, humidity) could also be used to further adjust the intensity of light when motion was detected. This ability to change the light response without continual parametric change meant the total amount of light used could be most reflective of the actual lighting need of the outer conditions and the real-time detection of motion. When combined, all the light and sensor units were coordinated via the ZigBee wireless standard to a single hub. Combining all outdoor lights into a decentralized mesh network of wireless poles also allowed sharing of sensor data and a coordinated response with respect to lighting.

A dynamic lighting approach to lighting designs meant that the energy consumed was actively managed and never wasted. The lighting could also have been scheduled historically based on time of day and average traffic trends, as long as there had been enough historical data about movements. The live deployment in an urban context was studied within a wide variation of environmental conditions and successfully met the challenge of balancing energy savings for the community with the local public safety light levels for both pedestrian and vehicular traffic. In real-world testing, ZigBee communication was found to transmit weaker signals, especially in urban settings with high density of electronic devices. This was a concern in terms of reliable systems, particularly for future urban deployments.

This work highlighted the value of merging a communication protocol with live sensing to both aid in awareness of energy utilization and the ability to make real-time energy

efficient decisions. It also provided a clear pathway to develop and integrate adaptive lighting systems for smarter urban functioning.

Ruchika Prasad [7] described a case study of IoT enabled smart street lighting the region of Nagpur. The purpose of the infrastructure was to inform sustainable energy options via the implementation of an energy efficient smart street lighting system empowering the local authority to collect data in real time posting installation and via the cloud continually monitor and manage street lighting dependent upon environmental parameters while significantly reducing energy usage and ongoing maintenance costs of energy lighting infrastructure. Ruchika Prasad completed a case study on an IoT-based smart street lighting solution launched in Nagpur, India. The investigation was focused on measuring how effective new technologies were transforming the traditional form of street lighting to more intelligent energy-efficient systems. The solution involved cloud computing, real-time sensor data, and centralized control to achieve improved efficiency and lower maintenance costs for urban lighting.

Every streetlight was installed with sensors to collect ambient light, temperature, and movement. That information was sent to a cloud platform, where the data was analyzed to create an optimal lighting pattern. The system was able to change brightness, set maintenance alerts, and even turn off when not needed. These possibilities provided energy savings and facilitated the repair of faulty lights if it was needed.

The project had substantial value in its ability to be remotely monitored and used predictive analytics. The System administrator had access to real-time lighting information, was notified if a fault occurred, and could manage energy consumption for the entire city from one dashboard. This improved operation and transparency and sophistication and accountability. While there were considerable costs associated with the investment in sensors and connectivity infrastructure, Prasad showed that the long-term savings related to energy and maintenance expenses outweighed the costs. The case study demonstrated how a city can develop its infrastructure to achieve smart city objectives.

In summary, Prasad's research indicated an effective implementation of IoT for public lighting in the real-world. It demonstrated the importance of data-informed decision making, and it provided an example of a model that was scalable and replicable, one which other cities could learn from and could use to improve their urban infrastructure.

Adnan M. Al-Smadi et al. [8] developed an energy-efficient smart street lighting system using IoT and microcontrollers. The smart street lighting systems have the ability to

monitor and detect faults remotely and help ensure minimal energy consumption via external human factors, as well. Adnan M. Al-Smadi et al. proposed an Internet of Things (IoT) based Smart Street Lighting System that aims to improve energy efficiency, create a remote monitoring application, and automate fault detection and diagnosis. The researchers designed a model using microcontrollers, light environmental sensors, and wireless communication modules, enabling a collection of smart street lights that can control and reduce downing (lighting output), and relay failures in real time.

The microcontroller processed data from motion sensors and LDR devices to make a determination on the street lights' needs. When the surrounding light decreased, or imposed motion was detected, the street lights auto-adjusted their brightness. When faults occurred (an operational failure due to lamp failure or other unusual power draw), faults triggered alerts that were transmitted to a website-based remote dashboard that informed maintenance personnel of a failure.

One clear contribution of this work was how well it found the sweet spot between energy savings and functional reliability by reducing the amount of time too much light was unnecessarily downing and enabling rapid response when a fault developed, which connected to two of cities' most stressed urban situations - usage of electricity, and street light down-time. Communication was achieved through a wireless module that implemented a good IoT model making it more scalable and better able to integrate into the cloud. Data was stored through and visualized through an online platform. This data provided an easy to understand interface in which the administrator could manage the public lighting infrastructure.

Although this design accomplished what we wanted and provided great success, it was highly internet dependent and its remote managed to rural or poorly infrastructured environments was potentially problematic. In the most general sense, the solution had great opportunity effectiveness and application in smart cities, based on low costs of hardware and useable nature of the design.

The paper was an important step in proving that IoT and automation could be applied practically in the public lighting space and demonstrated the relevance of fault detection in terms of energy optimization and not just smart city accountability.

Omkar Rudrawar et al. [9] presented a street lamp with light intensity control using power electronics. The method enabled adjusting LED light levels based on ambient light intensity thereby reducing energy consumption. Omkar and colleagues ultimately represented the benefits of using electronics with optimized control in their streetlight

project. Omkar Rudrawar et al. designed a smart street lighting control system that manipulated intensity levels via power electronics that measured the environment. Primarily, the central problem what their system solved was the constant power usage of traditional streetlights that turn on at full intensity without addressing need (that is, undimmed) or the surrounding environment.

Their system was primarily self-driven. It utilized sensors (LDRS, or light dependent resistors) to take readings of ambient light. Once the sensing was interfaced to a microcontroller, LED light could be varied by utilizing power electronics circuits (dimmers and regulators) to control light intensity. With this system could be set to dim during dawn, dusk, or even in conjunction with calibrated ambient light usage, or indeed turn itself on during the complete darkness of the night.

The authors demonstrated that their control system could effectively be deployed in urban and semi-urban scenarios. Unlike traditional systems that simply switch lights on or off, their system was capable of intensity level variances, which resulted in dynamic decrease in leveraged electricity usage. By focusing on using analog sensors combined with microcontrollers the authors also demonstrated economic feasibility for developing regions. There were many good aspects of this paper, including the ability to control the lights in real-time, which ensured that the lights functioned optimally without any human maintenance. The authors also included fault detection capability, whereby the system could notify when individual lamps were abnormally drawing power or could determine that a light was not functioning at all. While it was clear the authors were capable of building a notification capability into the IP network, the lack of notification capability tied into GSM or other cloud applications did limit the monitoring to more local, and potentially less useful, notifications.

Although the system was obviously scalable, the authors carried the concept of scalability into the operational model whereby, even if a centralized control system (or no central control system at all) was utilized, each lamp operated autonomously based on any light condition readings that occurred locally, so the solution is fault tolerant and can be deployed in as small of a quantity and duration as necessary.

In conclusion, the study accomplished its intended purpose in illustrating that, by combining environmental sensing and power electronics, a smart, responsive, and energy efficient street lighting system could be created. The authors provided a definitive hardware-focused solution for smart city lighting solutions, but as a note for future directions, basic IoT functionality and centralized notification and monitoring would provide

even greater value to communities.

Gourab Das [10] proposed a predictive model used to optimize streetlight intensity utilizing artificial intelligence. The proposed solution processes historical traffic data to predict streetlight lighting requirements for energy regulation whilst ensuring safety in people that are active in the traffic environments. Gourab Das devised a predictive AI-based model for streetlight intensity optimization. Unlike old-fashioned systems that adapt to changing levels of motion, this model used historical and verified data of traffic levels and patterns to forecast what amount of light would be required, maintaining energy efficiency and maintaining road safety.

The model used machine learning algorithms to learn about recorded historical traffic flow and historical environmental conditions (weather, daytime, day of the week), in the process of determining optimal lighting levels. Based on predictions, the system was capable of proactively brightening or dimming streetlights, based on forecasts of how dense traffic was expected to be, as opposed to traditional systems that only respond when motion was detected.

Data pre-processing, and model training were key components of the system. Gourab's model used supervised learning methods, including regression and time-series forecasting methods, to assist in making intelligence predictions. The model was then deployed into a simulated environment to results to assess model accuracy and impact on energy savings. In terms of the predictive model's efficiency, this was clearly higher than PIR-based or LDR-based systems, as they did not switch unnecessarily and had less risk of abrupt changes to lighting levels. Data availability and model drift were quite explicitly cited as limitations, and the reduced accuracy was acknowledged over time until retrained.

The infrastructure requirements were a significant step above traditional systems, requiring access to near real-time traffic feeds, and had to run in a cloud computing environment that was well defined and reliable. However, the potential for integration with smart traffic management platforms is a compelling argument for municipalities when creating their urban planning scope.

This work represented a clear step forward in the direction of AI driven infrastructure, and offered an example of how intelligent analytics can serve to improve energy usage policies, and furthermore alter lighting to respond dynamically to urban conditions. Gourab's method proposed a model for urban planners, which could be useful in terms of creating some level of integration into the growing trend of smart cities, where predictive maintenance and automated adjustments are all foundational pieces of

a scalable and repeatable infrastructure.

Priyanshi Rami et al. [11] presented an intelligent streetlight controller that was simulated to energy savings based on traffic flow. Using MATLAB to model results, their simulation model indicated a significant overall reduction in power would be used in low traffic scenarios when streetlights adapt to less intense functionality. Priyanshi Rami and coworkers showed an intelligent streetlight controller that adjusts light brightness in response to traffic. The tests were carried out almost exclusively using MATLAB simulations of traffic flow and energy consumption.

Their smart lighting system architecture consisted of motion sensors, microcontrollers, and LED lights. The motion sensors detected the movement of vehicles or pedestrians; the microcontroller then interpreted this data with logic to turn on or dim the street lights. In essence, instead of always being lit, the lights switched on, or dimmed according to actual traffic activity, saving on electricity and prolonging the life of the LED lights.

The research was very interesting because of the simulation model developed in MATLAB. The authors used real traffic flow data to show the potential of smart lighting to significantly reduce power use during low-traffic situations. The authors concluded with an assessment of their findings. They indicated that even if only part of the smart lighting system were to be implemented in a city, it would still result in savings on energy costs, and those savings would easily be seen by local authorities. While the results were encouraging, the paper emphasized that the actual use of the technology would require integration with communication protocols, a hardware redundancy plan for handling sensor failure, and other considerations. In addition, there were still challenges to moving from a simulation to the real world, including sensor calibration, environmental noise, and hardware durability.

However, the work was still valuable for validating theoretical energy savings and for demonstrating a pathway towards hardware prototyping. The system could also be enhanced by the inclusion of environmental sensing or AI prediction models for greater operational optimization in the future.

In conclusion, the current work and research illustrated a simulation-based platform for smart street lighting, demonstrating possibilities for dynamic light control systems responsive to traffic conditions. The work provided a solid path forward for potential real-world implementations in urban traffic control settings.

Mohsen Mahoor et al. [12] produced a comprehensive review of smart streetlight and

throughout they discussed the evolution of streetlight lighting based on sensor data, machine learning, and artificial intelligence. They pointed to some continuing problems and challenges with the implementation and compatibility of systems, real-time data that provides appropriate analytics. The authors, Mohsen Mahoor et. al., conducted a state-of-the-art review on smart street lighting systems, which included discussion of the evolution of the technology and integration of sensors, machine learning, and AI towards the use of smart public lighting. The review included many applications (research and pilot) and comparisons among the technologies and architectures used, as well as the mode of communication used.

One key theme that was reported in the review was the shift away from fixed-time (on/off) street lighting systems to intelligent lighting systems that use smart controllers based on data as the basis for the system. The authors emphasized that IoT based streetlights are equipped with environmental and motion sensors to control brightness, detect faults, and even consider lighting needs using AI models.

The review also compared different communication protocols (ZigBee, Wi-Fi, GSM, and newer LPWAN method systems like LoRa, Wi-SUN), and weighed their attributes; bandwidth, operating energy, and immunity to interference. The review also covered cloud computing and edge computing systems for processing, storage, and analytics of all the data derived from the sensors. One of the significant contributions of this paper was the solid overview of some current issues. These issues included elevated installation and maintenance costs, a lack of interoperability, sensor reliability in poor weather conditions, and data security vulnerabilities. The paper included another noted bottleneck to large-scale deployment related to the lack of standardized protocols and scalable models.

Mahoor et al. outlined possible future research directions, indicating that more hybrid AI-IoT, edge analytics and cloud-based insight models were viable next steps. The authors also suggested more robust testing in varied environments to enhance reliability and performance.

This review can inform significantly to academics and engineers engaged in the smart city domain, presenting a tangible overview of the state-of-the-art while highlighting the challenges and research gaps experienced in the smart street lighting space.

Sampath Kumar Pappula et al. [13] presented the idea for an IoT based auto intensity controller with fault detection, and light dependent resistors (LDRs). The system was found to automatically control the brightness and detected failure sending the closure to cloud-based platforms. Sampath Kumar Pappula et al. proposed an IoT-based auto-

intensity streetlight controller with a fault detection system. The main aim of their study was for improved energy efficiency and little human interaction in the management of public lighting infrastructure. The authors utilized Light Dependent Resistors (LDRs) and microcontrollers to adjust light intensity depending on ambient light, and to detect faults in the lighting system.

The system architecture included LDR sensors to detect light intensity, microcontrollers, such as Arduino or NodeMCU, and LED streetlights. Microcontrollers used real-time LDR readings to modulate the brightness of the LEDs using Pulse Width Modulation (PWM). The system would leave the lights at full brightness at night, but would dim the lights to a certain intensity at dawn or while it was overcast. The system also had fault detection via current sensing, to indicate defective or non-functioning lights.

The novelty also further extended to the IoT component of the project: the microcontroller was capable of sending data about faults and lightning status to a cloud-based platform via Wi-Fi (ESP8266 module). This enabled the eventual application of the controller to be monitored remotely by municipal authorities who could monitor the status of each streetlight remotely on a dashboard or mobile application in real-time. The paper presented an effective and expandable solution for intelligent street lighting that is readily transferable to urban and suburban areas. The link together utilization of real-time monitoring, automatic intensity adjustment and fault reporting and alerts in a relatively inexpensive model are effective and maintain sustainability. The added benefit of using available IoT technologies and communications protocols made the solution consistent with a smart city concept.

The authors also pointed to some limitations. There may be variables outside the control of the system, such as environmental conditions that affect sensor reading (i.e., foggy and rainy conditions). Reliability of Wi-Fi in dense or remote environments could diminish the freedom to transfer data in real-time. The authors pointed to improvements they would make next, such as using LoRa (for longer range communication) and with AI to incorporate predictive fault analytics in future implementations.

In summary, the paper described a method for control of automated intelligent streetlight, which enabled automation, energy savings and real-time fault detection through the integration of IoT technologies. The authors provided a reliable prototype for future consideration of smart urban denizenship.

Yuxi Jiang et al. [14] created a street lighting system using wireless technology based on real-time traffic sensing. It used sensors, so the light levels could be adjusted instant-

neously for all areas, saving energy while also providing safety lighting, regardless if there was traffic or not. Yuxi Jiang and colleagues created an intelligent street lighting system that utilized a real-time traffic sensing system and wireless communication system. Their main aim was to use as little energy as possible while adjusting the illumination of street-light luminaires in real-time based on the flow of traffic and the amount of traffic. The project was in support of a larger initiative to develop responsive and sustainable smart city infrastructure.

The street lighting system has several primary components: traffic sensors (e.g., ultrasonic or IR sensors), microcontroller, and a wireless communication module (ZigBee or Wi-Fi). The sensors were placed in several locations around and along roads to monitor the presence of vehicles, speed of vehicles, and density of vehicles. When there was detected traffic, the corresponding luminaires would increase illumination; otherwise, the luminaires would decrease their illumination, or not turn on, which would ultimately save energy during the times with little or no traffic.

The main highlight about the study was that the authors used a real-time resonant lighting system. The authors highlighted that their lighting system provided electrical illumination only while sensing traffic, therefore, streets were lit only when required, without compromising the safety of pedestrians and vehicles. The authors' response and on-demand illumination reduced electric lighting use without the cumbersome requirement of a set schedule of lighting, and limited power based on ethical considerations of safety from motion, or even light delays. Furthermore, the system enabled centralized monitoring. Sensor information was transferred wirelessly to a control center so that patterns could be analyzed, reports created and the system could be modified by remote control. The communication backbone of the solution allowed for future growth and integration with other smart infrastructure, including traffic signals and environmental monitoring stations.

Yuxi Jiang's system did offer significant advantages over fixed-time or PIR-based systems. However, the researchers expressed some concerns over continuous wireless communication, especially because in an urban environment buildings and traffic can interfere with signal transmission. The other consideration was the capital costs because of such dense sensor networks and the requirements for installation and maintaining the communication networks.

Overall, the paper provided a valuable contribution to the field of intelligent street lighting by providing a real-time, responsive approach to traffic. The system provided a

good balance between energy efficiency, safety, and automation and established a starting point for integrated traffic-lighting management systems.

Tejas Deshpande et al. [15] developed a solar powered LED lighting system that included digital dimming control. Their innovative approach was to have Light Dependent Resistors (LDRs) in conjunction with motion sensors to optimize light levels. It was also solar powered, which contributes to being the most sustainable while at the same time cost effective. Tejas Deshpande and his team designed smart LED street light utilizing solar power, which incorporated digital dimming and smart brightness capabilities that could adjust based on motion and ambient light intensity. Their plan revolved around combining sustainability with automation by utilizing solar power with LDR and PIR (Passive Infrared) sensors in an effort to reduce their power consumption and reliance on grid electricity. The system architecture for the street light would be comprised of a solar panel connected to a battery system, which powered the LED streetlights. An LDR sensor would detect ambient light intensity (to determine whether it was day or night) and a PIR sensor would detect movement in the area. When there is low ambient light (i.e., during the night) the lights would remain dim until it is detected that movement occurred, at which point the light would illuminate at full light for a set period. What was unique about the design, and what made the authors proud, was the design's attention towards energy autonomy. Using the solar panel to power the street light with battery storage would allow the street light to remain independent from conventional power sources altogether, making it the most effective option for rural locations or off-grid circumstances. The combination of digital dimming control and motion/light sensors would allow for maximum battery sleep duration. If the streetlight was adequate, it could run for several nights without getting sufficient solar input below the threshold. The authors provided a microcontroller to process sensor inputs and light control logic; therefore, improving responsiveness of the original system. By providing a reasonable way to dim the lights through PWM signals, the system could change brightness fairly quickly for decent savings without a second delay or detectible flicker.

Field testing results showed quite a bit of promise, showing major decreases in required power usage and maximum battery capacity discharge. The system supports energy savings while reasonably balancing reliable light when needed, by lighting the streets as described. In addition the LED's use isn't full intensity AND continuous means that extends their life too.

Some certain limitations arose in that the setup system (including the solar panel and

battery systems) is large and leads to larger costs in the initial setup, and not as effective in areas with large cloud periods. But, they made recommendations for hybrids systems (grid + solar) or bigger batteries.

Overall, these studies reflect a robust shift towards intelligent, connected, and sustainable streetlight solutions. Trends in leveraging IoT, renewable energy, image processing, and communication protocols (ZigBee, Wi-SUN) offer innovative pathways to address issues of energy waste, maintenance, and scalability. However, challenges remain in cost optimization, sensor accuracy, and long-term reliability in varying environmental conditions. Future research can focus on hybrid models that combine AI, edge computing, and real-time analytics for smarter, self-diagnosing streetlight networks.

Chapter 3

Problem Statement

Streetlights are critical for safety and fluid mobility on urban streets after dark. Traditional street lighting systems suffer from drawbacks: no real-time fault detection, slow response to maintenance, and poor energy management. All of these can lead to two or three week outages, more operational cost, and risks for public safety.

Currently, the maintenance is reactive, relying completely on manual inspections and or scheduled monthly inspections which can take a lot of time, or quite simply, just are not efficient, especially on large dense urban environments. As urban areas become smarter and more sustainable, fast, smart solutions are urgently needed to monitor streetlights continuously, detect faults quickly and notify the appropriate authorities through an automated system.

This project aims to address these issues by proposing a Smart Streetlight Fault Detection and Monitoring System that enables proactive maintenance, improves operational efficiency, and enhances the reliability of urban lighting networks.

Chapter 4

Requirements Specification

Software Requirements

Software	Version	Speed/Performance	Size	Variants	Quantity
Arduino IDE	1.8.19 or later	Fast compile/upload on USB	~150 MB	Windows / Mac / Linux	1
Python	3.11 or later	High scripting speed	~30–40 MB	CPython, Anaconda	1
Firebase	Latest (Web)	Real-time data streaming	Cloud-based	Web dashboard / API	1
MQTT	NA	Lightweight messaging protocol	Few MB (library)	Mosquitto, HiveMQ	1

Table 4.1: Software Requirements

Use of Software Components:

- **Arduino IDE:** Used for writing, compiling, and uploading code to microcontrollers like NodeMCU.
- **Python:** Used for backend processing, data manipulation, and integration with APIs.
- **Firebase:** Provides a real-time database to store and sync sensor data in the cloud.
- **MQTT:** Enables lightweight, efficient communication between IoT devices and servers.

Hardware Requirements

Component	Version	Speed	Size	Variants	Quantity
NodeMCU	ESP8266	80/160 MHz MCU	Compact (2.4")	ESP-01, ESP-12E, ESP32	1
LDR Sensor	NA	NA	Small (PCB mount)	Analog / Digital types	3–5
GSM Module	SIM800L	Quad-band GSM	Small (2G module)	SIM900, SIM800	1
GPS Module	NEO-6M	9600 bps	1.5" × 1.2"	NEO-M8N, NEO-7M	1
Jumper Wires	Male-Male	NA	20–30 cm	Male / Female	1 set
Power Supply	5V DC	2A	Wall adapter	USB, Barrel plug	1

Table 4.2: Hardware Requirements

Use of Hardware Components:

- **NodeMCU (ESP8266):** Acts as the main controller for processing and transmitting data over Wi-Fi.
- **LDR Sensor:** Detects ambient light levels; useful in automation and monitoring systems.
- **GSM Module:** Enables SMS-based communication when internet access is unavailable.
- **GPS Module:** Provides location data, useful in tracking and navigation applications.
- **Jumper Wires:** Connect various components on a breadboard or PCB without soldering.
- **Power Supply:** Provides the necessary electrical power to the microcontroller and sensors.

Chapter 5

Proposed Methodology

5.1 Overview

The proposed system aims to automate the detection and reporting of faults in streetlight infrastructure. It integrates hardware sensors with a cloud-based software platform that enables real-time monitoring, notification, and management of repair tasks.

5.2 System Architecture

The system architecture is divided into three main components:

1. **Sensing Layer:** Includes hardware sensors and microcontrollers that detect faults (e.g., lights not turning on, broken components) and environmental parameters (e.g. motion, ambient light).
2. **Communication Layer:** Uses a GSM module (MQTT in alternative setups) to transmit fault data to the central server.
3. **Application Layer:** A web-based interface for administrators and maintenance workers to monitor, track, and manage streetlight repair tasks.

5.3 Step-by-Step Methodology

1. **Sensor Deployment:** Each streetlight is equipped with sensors to detect operational status (e.g., voltage, current, light intensity).

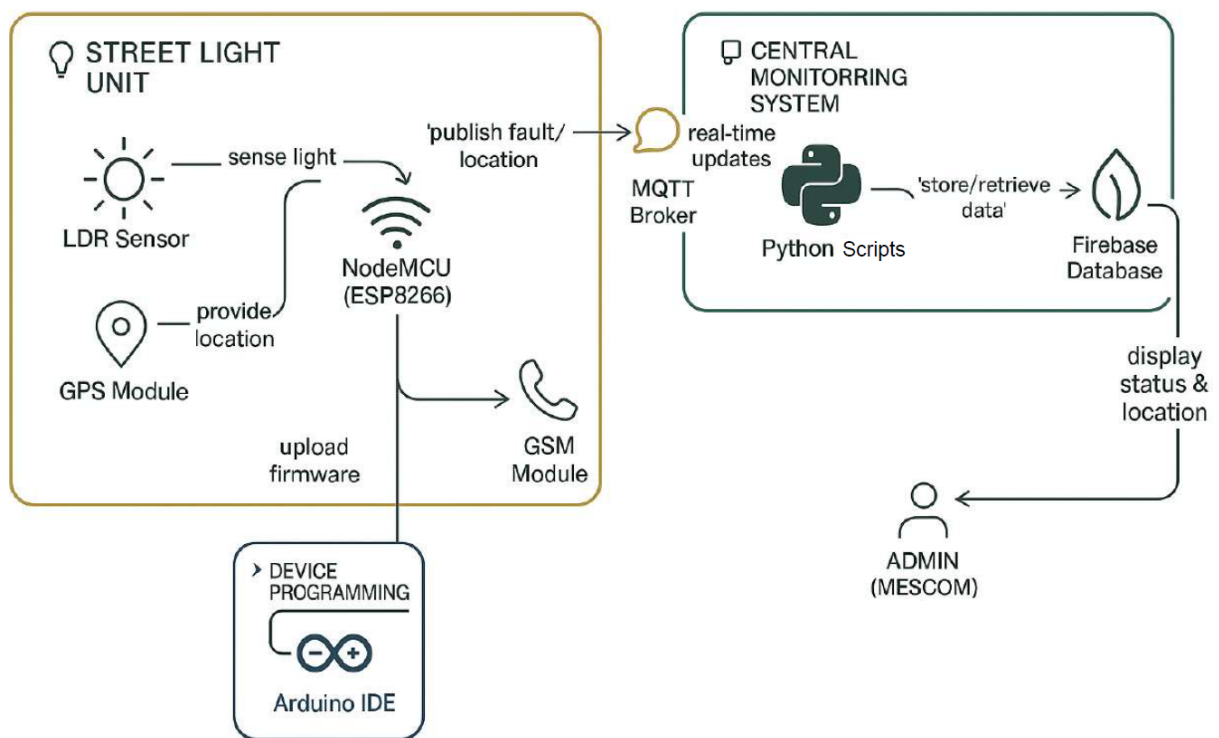


Figure 5.1: System architecture of the smart street lighting system

2. **Fault Detection:** The microcontroller continuously monitors the sensor data. If anomalies are detected (e.g., zero voltage during night hours), a fault is logged.
3. **Alert Transmission:** Fault data is sent via the GSM module to a central server or cloud-based system.
4. **Data Logging:** The server receives and stores the fault data into a structured database.
5. **Web Dashboard Interface:**
 - **Admin Panel:** Displays real-time faults on a dashboard with location, type, and severity.
 - **Task Assignment:** Admin assigns repair tasks to available workers based on location and workload.
6. **Worker Notification:** The assigned maintenance worker receives the task via the web/mobile interface.
7. **Repair and Status Update:** After completing the repair, the worker marks the task as completed. The system updates the fault status in the database.

5.4 Advantages of the Proposed Methodology

- Reduces manual inspection time and cost.
- Enables faster repair and better resource allocation.
- Improves public safety and system efficiency.
- Scalable for integration into smart city infrastructure.

Chapter 6

Expected Outcomes

At the successful completion of the project titled **"Monitoring System for Street Light Fault Detection And Location Tracking"**, the following outcomes are expected:

1. **Real-time fault detection and reporting**

Design and implement smart street light systems which use sensors and micro-controller technology to detect faults, such as lamp failures and anomalous data (current, voltage), and relay the fault data on a real-time basis to a centralized server using GSM or WI-FI modules, which collects data such as the street light ID, type of fault, time (datetime) of detection. The real-time data collection allows for the monitoring of street light functionality without manual inspection tasks, thus making maintenance activities more efficient.

2. **Centralized monitoring and task management**

Develop web-based dashboard for administrators to monitor the status of all street-lights and their live operational status. The dashboard will allow the administrator to assign tasks based on a technician's availability and location of faults. This central monitoring and task assignment tool will record and link the tasks to fault records and technician logs, allowing for real-time updates to task assignments, status of completion, as well as accountability for both technicians and the administrator in regards to task scheduling and resource management.

3. **Technician interaction and prototype validation**

The dashboard interface will allow maintenance technicians to update the status of repairs made to each street light (fault repaired), which will provide real-time notifications to the administrator, and provide for detailed and accurately maintained

records of maintenance activity. The on-site prototype testing will be a small-scale implementation to evaluate the full workflow process from fault detection to fault repair confirmation and accurate logging, and to validate the overall functionality and practical applicability of the smart street light systems.

Chapter 7

Project Plan

The project "Monitoring System for Street Light Fault Detection And Location Tracking" will be developed in several simple and clear steps. Each step helps us build and test the system step-by-step.

Step 1: Understanding the Requirements

In this step, we discuss and understand what our system should do. We list all the parts we need, like sensors, microcontrollers, and lights. We also decide what software and tools we will use.

Step 2: Designing the System

Here, we plan how everything will work. We draw diagrams to show how the parts connect and how data will flow. We decide how the sensors, lights, and other components will interact.

Step 3: Setting Up the Hardware

We connect all the electronic components like sensors, microcontroller, and LEDs. We make sure that each part works properly. We also test if the microcontroller can read sensor values.

Step 4: Writing the Code

Now, we write the program that runs on the microcontroller. The code will read sensor data and check for faults in the lights. It will send the information to a cloud server or database for monitoring.

Step 5: Testing the System

After connecting hardware and software, we test the whole system to make sure it works correctly. If something doesn't work, we fix the problem. We also check if the data is being displayed properly.

Step 6: Final Integration and Demo

Once everything works well, we put all parts together and test the full system. We show how the system can detect faulty streetlights and send the data for remote monitoring.

PROJECT WORK FLOW											
TASK (MONTHS)	2024-25 EVEN						2024-25 - ODD				
	FEB.	MAR.	APR.	MAY.	JUN.	JUL.	AUG.	SEP.	OCT.	NOV.	DEC.
SDG & SUB-DOMAIN IDENTIFICATION											
TOPIC SELECTION											
SYNOPSIS PREPARATION											
PRE-PHASE – I PRESENTATION (IDEA PRESENTATION)											
LITERATURE REVIEW											
PHASE – I PRESENTATION (LR, SRS, ARCHITECTURE)											
PROJECT DESIGN											
PROJECT IMPLEMENTATION											
PRE-PHASE – II PRESENTATION (DESIGN & IMPLEMENTATION PRESENTATION)											
PROJECT TESTING											
PHASE – II PRESENTATION (COMPLETE DEMONSTRATION OF THE PROJECT)											
REPORT WRITING											
PUBLICATIONS/PATENT/PRODUCT DEVELOPMENT											

Figure 7.1: Gantt Chart

Chapter 8

Conclusion

The Smart Streetlight Fault Detection and Monitoring System expounded on throughout this project exhibits a strong, effective, and forward-thinking answer to the limitations of traditional street lighting infrastructure. By utilizing an embedded system with components such as a NodeMCU microcontroller, multiple environmental and electrical sensors, and wireless communication protocols utilizing MQTT over Wi-Fi, the system enables real-time fault detection, automated data communication, and remote monitoring. This smart solution targets typical urban infrastructure weaknesses, such as delayed fault resolution, high energy usage, and no visibility of public utility operations.

A fundamental strength of the system presented is an inherent ability to autonomously detect fault conditions, such as lamp malfunctions, current anomalies, drops in voltage, or disconnections. The sensor nodes detect faults whether through a lamp failure that stops all current operations, a current anomaly through the current sensor, or a lack of power through a drop in voltage level, which also stops all current operations. Regardless of how it is detected, faults must be reported as soon as possible. By utilizing a log as part of the built-in processes in the NodeMCU, the sensor node reports faults to the web application via MQTT in as little as 1-2 seconds. Once the web application receives the information, reports, including location of faults (potentially utilizing GPS coordinates), real-time system health, past fault, and more are available to the administrator in real-time on their dashboard. Overall, if implemented successfully, an infrastructure of smart streetlights could mitigate costs associated with faults, provide better accountability for faults, enable better maintenance planning (based on fault history), and prevent the transport of technicians for scheduled visits that may not be of a high certainty. Moreover, the system provides an adaptive lighting control capability, based on ambient light as

well as vehicle/pedestrian presence. This capability allows streetlights to dim or brighten according to conditions or presence, thus saving energy usage in a great deal. Dimming or brightening avoids wasting energy and also operates lighting components for only as much as is needed, i.e. the adaptive lighting capability can all but double the life of a streetlight through an effective adaptive management approach.

From a software architecture perspective, this system provides control dashboards for two separate user roles, the administrator and the field technician. The administrator can visualise the locations of faults, assign repairs, and visualise technician responsiveness. For the worker or technician, the use is quite simple, as the interface provides realtime updateability, progress acknowledgement on faults, and post-repair verification. Overall, these features create a better life cycle for maintaining streetlights, as it creates a close-loop feedback system for operations and repairs. It allows for faster and clearer accountability, while also reducing delays of manual records. The project also enhances sustainability and cost-efficiency. The reduction of energy usage, manual labor, and expensive extended outages, provide reduced operational costs and a lower carbon footprint, which all favor sustainable workplace initiatives from local, national, and international perspectives (with regards to national and international policy and directions), such as sustainable cities and clean energy in relation to the United Nations Sustainable Development Goals (SDGs).

In summary, this project represents a significant example of how embedded systems, IoT, and logic control algorithms can be merged to improve the delivery of urban infrastructure. The Smart Streetlight Fault Detection and Monitoring System represents not only a technological innovation but an initiative towards building safer, greener, and smarter cities. If further enhancements are made, such as solar capabilities, voice assistant capability and AI-based analytics, the Smart Streetlight Fault Detection and Monitoring System could become a part of future public infrastructure.

References

- [1] A. Quattrocchi, F. Martella, V. Lukaj, R. De Leo, M. Villari, and R. Montanini, “Designing a Low-Cost System to Monitor the Structural Behavior of Street Lighting Poles in Smart Cities,” **Sensors**, vol. 23, p. 6993, 2023.
- [2] E. Dizon and B. Pranggono, “Smart streetlights in Smart City: A case study of Sheffield,” **Journal of Ambient Intelligence and Humanized Computing**, 2022.
- [3] A. Omar et al., “Smart city: Recent advances in intelligent street lighting systems based on IoT,” **Journal of Sensors**, 2022.
- [4] M. F. Younis and S. S. Salim, “Cloud-based automatic street lighting control system,” **Journal of Physics: Conference Series**, vol. 1963, 2021.
- [5] S. Gong, R. Kumar, and D. Kumutha, “Design of lighting intelligent control system based on OpenCV image processing,” **International Journal of Uncertainty, Fuzziness and Knowledge-Based Systems**, vol. 29, Supp01, pp. 119–139, 2021.
- [6] M. Sahithi Prasanthi et al., “IoT Based Streetlight Management,” 2018 3rd IEEE International Conference on Recent Trends in Electronics, Information Communication Technology (RTEICT), 2018.
- [7] N. Yashaswini et al., “Automatic street light control by detecting vehicle movement,” 2018 3rd IEEE International Conference on Recent Trends in Electronics, Information Communication Technology (RTEICT), 2018.
- [8] E. I. Archibong, S. Ozuomba, and E. Ekott, “IoT-based solar powered street light with anti-vandalisation mechanism,” 2020 IEEE International Conference on Mathematics, Computer Engineering and Computer Science (ICMCECS), 2020.
- [9] D. R. Khade et al., “Intensity controller of LED street lights,” 2017 International Conference on Circuit, Power and Computing Technologies (ICCPCT), IEEE, 2017.

- [10] M. A. K. Riyaz et al., “Graphene coated LED based automatic street lighting system using Arduino,” 2017 IEEE International Conference on Power, Control, Signals and Instrumentation Engineering (ICPCSI), 2017.
- [11] C. H. Prasad et al., “Auto Intensity Control of a Street Light by using PV cell,” *International Journal of Engineering Applied Science and Technology*, vol. 4, 2017.
- [12] H. A. Attia, A. Omar, and M. Takruri, “Design of decentralized street LED light dimming system,” 2016 5th International Conference on Electronic Devices, Systems and Applications (ICEDSA), IEEE, 2016.
- [13] A. R. R. Yaye, B. Gueye, and C. S. A. R. R., “Performance analysis of a smart street lighting application using LoRa WAN,” 2019 International Conference on Advanced Communication Technologies and Networking (CommNet), IEEE, 2019.
- [14] P. T. V. Bhuvaneswari et al., “LoRa based smart street light management framework,” *International Journal of Research in Electronics and Computer Engineering*, vol. 6, 2018.
- [15] E. I. Archibong, S. Ozuomba, and E. Ekott, “IoT-based solar powered street light with anti-vandalisation mechanism,” 2020 IEEE International Conference on Mathematics, Computer Engineering and Computer Science (ICMCECS), 2020.